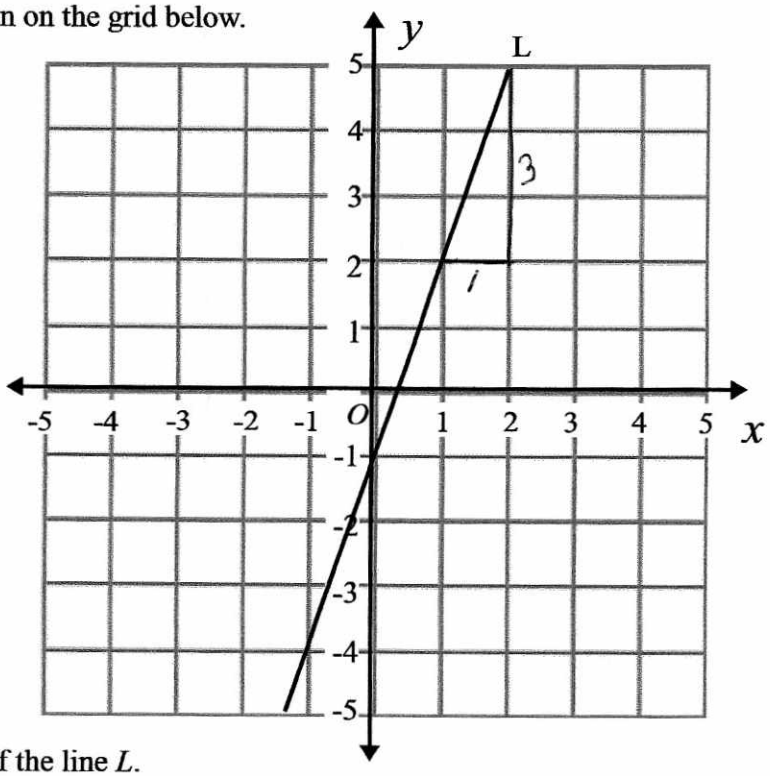
1 The line L is drawn on the grid below.

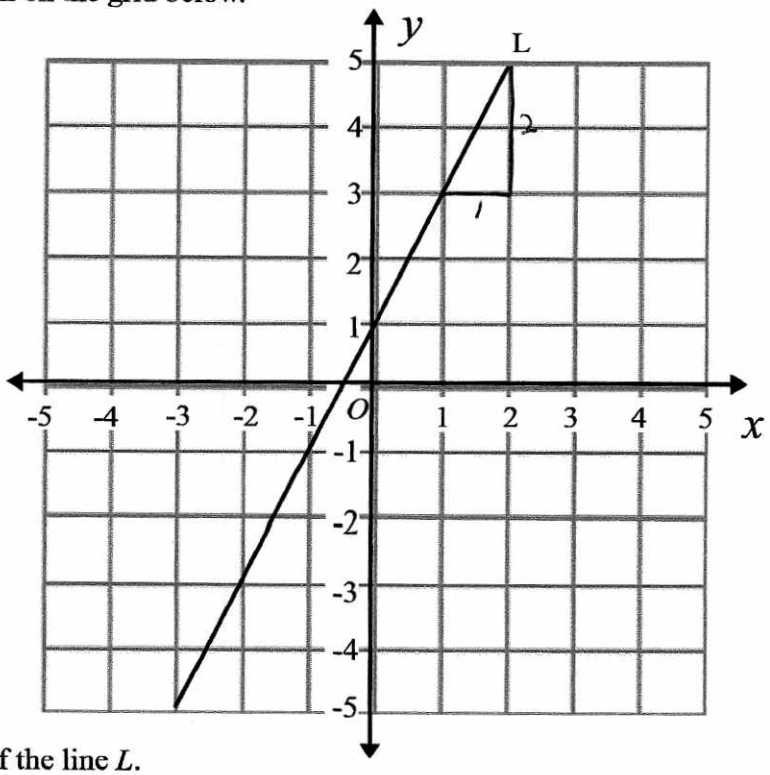


Find the gradient of the line L.

3

(Total for question 1 is 1 mark)

2 The line L is drawn on the grid below.

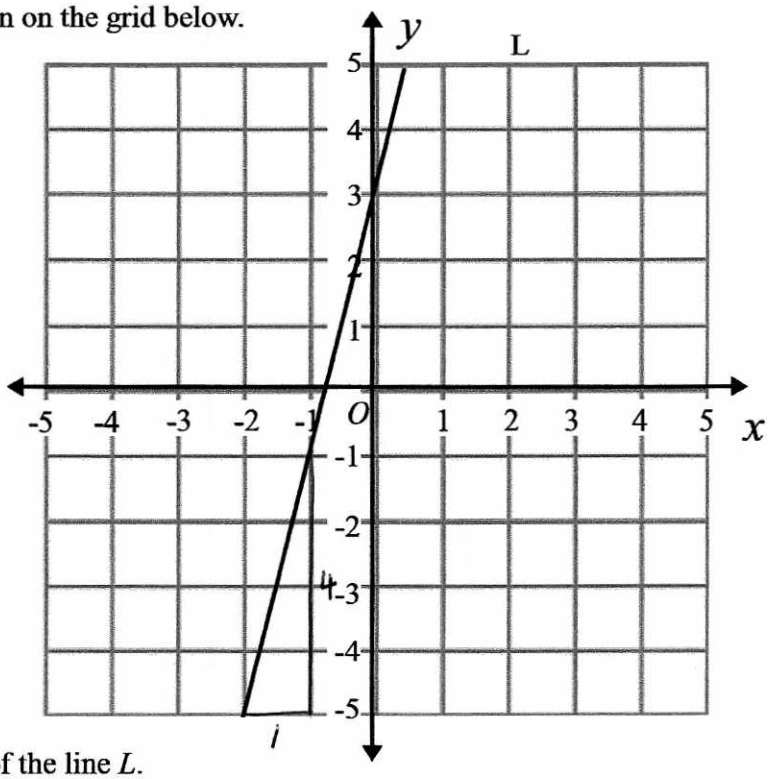


Find the gradient of the line L.

2

(Total for question 2 is 1 mark)

3 The line L is drawn on the grid below.

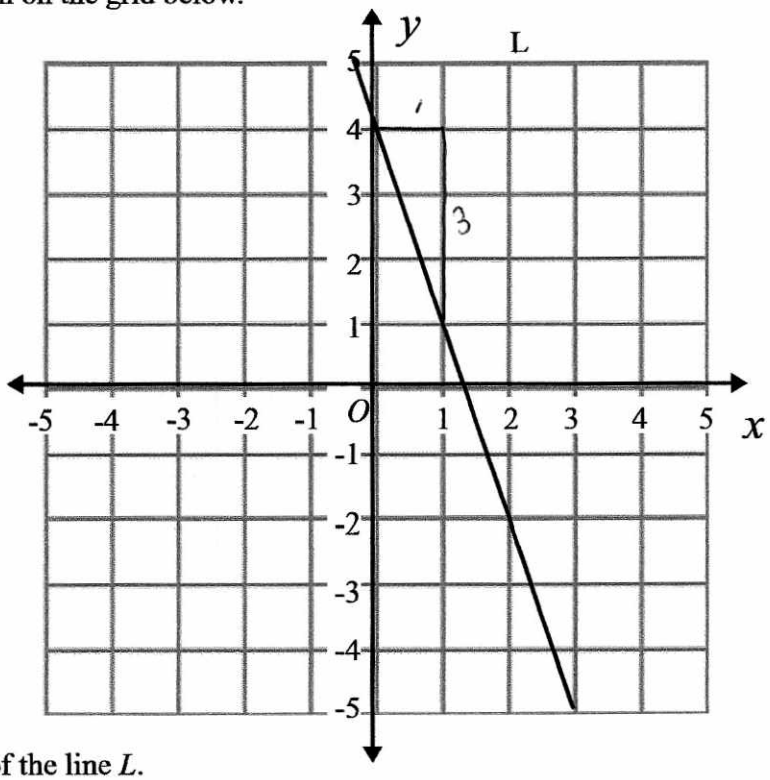


Find the gradient of the line L.

4

(Total for question 3 is 1 mark)

4 The line L is drawn on the grid below.

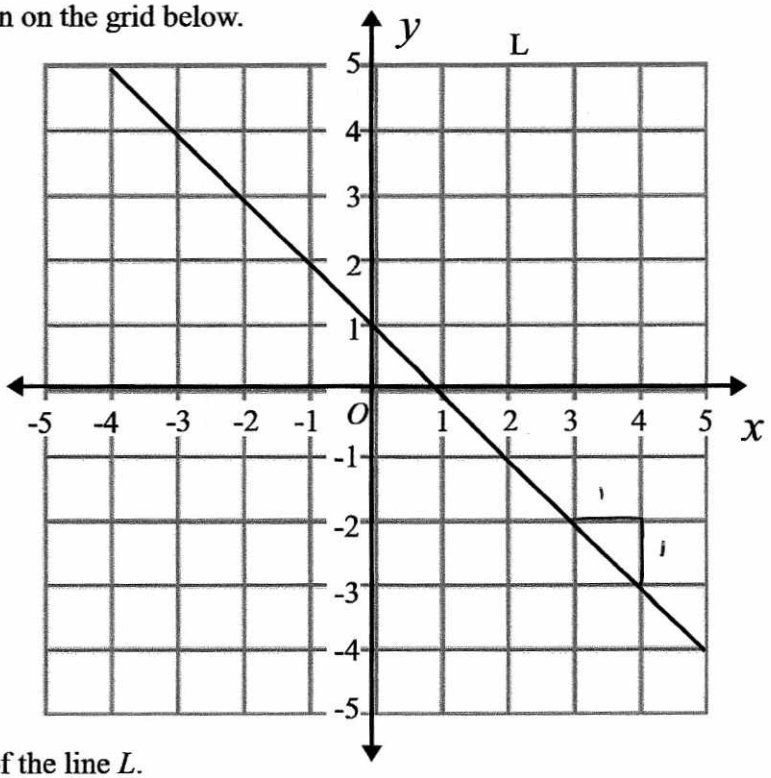


Find the gradient of the line L.

−3

(Total for question 4 is 1 mark)

5 The line L is drawn on the grid below.

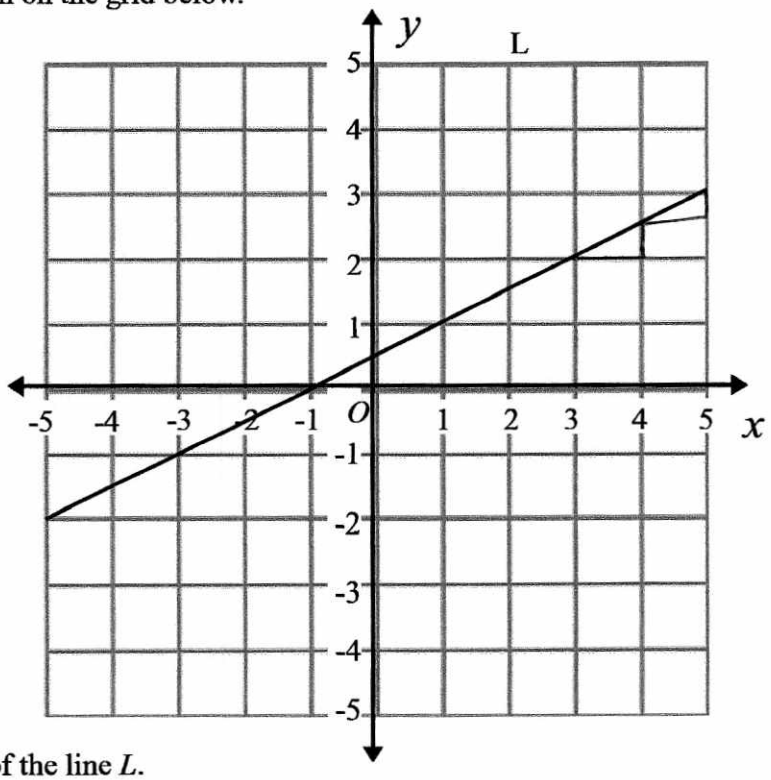


Find the gradient of the line L.

-1

(Total for question 5 is 1 mark)

6 The line L is drawn on the grid below.



Find the gradient of the line L.

$\frac{1}{2}$

(Total for question 6 is 1 mark)

7 Find the gradient of the line that passes through (2, 1) and (5, 10).

$x_1 \; y_1 \quad x_2 \; y_2$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 1}{5 - 2}$$

$$= \frac{9}{3}$$

3

(Total for question 7 is 2 marks)

8 Find the gradient of the line that passes through (5, 4) and (7, 0).

$x_1 \; y_1 \quad x_2 \; y_2$

$$m = \frac{0 - 4}{7 - 5}$$

$$= \frac{-4}{2}$$

$$= -2$$

-2

(Total for question 8 is 2 marks)

9 Find the gradient of the line that passes through (-3, 4) and (5, 8).

$x_1 \; y_1 \quad x_2 \; y_2$

$$m = \frac{8 - 4}{5 - -3}$$

$$= \frac{4}{8}$$

$$= \frac{1}{2}$$

$\frac{1}{2}$

(Total for question 9 is 2 marks)

10 Find the gradient of the line that passes through (3, 7) and (1, 10).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 7}{1 - 3}$$

$$= \frac{3}{-2}$$

$$= -1.5$$

-1.5

(Total for question 10 is 2 marks)

11 Find the gradient of the line that passes through (1, -1) and (-3, -9).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{-9 - -1}{-3 - 1}$$

$$= \frac{-8}{-4}$$

$$= 2$$

2

(Total for question 11 is 2 marks)

12 Find the gradient of the line that passes through (8, 1) and (3, -3).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{-3 - 1}{3 - 8}$$

$$= \frac{-4}{-5}$$

$$= \frac{4}{5}$$

$\frac{4}{5}$

(Total for question 12 is 2 marks)

13 Find the gradient of the line that passes through (3, -1) and (-2, 9).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{9 - -1}{-2 - 3}$$

$$= \frac{10}{-5}$$

$$= -2$$

-2

(Total for question 13 is 2 marks)

---

14 Find the gradient of the line that passes through (-1, -2) and (-3, 10).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{10 - -2}{-3 - -1}$$

$$= \frac{12}{-2}$$

$$= -6$$

-6

(Total for question 14 is 2 marks)

---

15 Find the gradient of the line that passes through (-3, 4) and (-5, 7).

$x_1 \quad y_1 \quad x_2 \quad y_2$

$$m = \frac{7 - 4}{-5 - -3}$$

$$= \frac{3}{-2}$$

$$= -1.5$$

-1.5

(Total for question 15 is 2 marks)

16 The line AB passes through the points A(2, -1) and (6, k).

The gradient of AB is 5.

$x_1 \quad y_1 \quad x_2 \quad y_2$

Work out the value of k.

$$5 = \frac{k - -1}{6 - 2}$$

$$5 = \frac{k + 1}{4}$$

$$20 = k + 1$$

$$k = 19$$

$k =$ 19

(Total for question 16 is 3 marks)

---

17 The line AB passes through the points A(-3, 4) and (k, 12).

The gradient of AB is 4.

$x_1 \quad y_2 \quad x_2 \quad y_2$

Work out the value of k.

$$4 = \frac{12 - 4}{k - -3}$$

$$4 = \frac{8}{k + 3}$$

$$4(k + 3) = 8$$

$$k + 3 = 2$$

$$k = -1$$

$k =$ −1

(Total for question 17 is 3 marks)

---

18 The line AB passes through the points A(-2, k) and (4, 8).

The gradient of AB is -2.

$x_1 \quad y_1 \quad x_2 \quad y_2$

Work out the value of k.

$$-2 = \frac{8 - k}{4 - -2}$$

$$-2 = \frac{8 - k}{6}$$

$$-12 = 8 - k$$

$$-12 + k = 8$$

$$k = 20$$

$k =$ 20

(Total for question 18 is 3 marks)